

CLASS TEST - 3

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① (a) $x = r \sin \theta \sec \phi$; $y = r \cos \theta \sec \phi$; $z = r \tan \phi$

$$\frac{\partial}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi}$$

$$\therefore \frac{\partial r}{\partial x} \cdot r = \sqrt{x^2 + y^2 + z^2} \quad ; \quad \tan \theta = \frac{x}{y} \quad \text{--- (i)}$$

Differentiating (i):

$$\tan \phi = \frac{z}{r} \quad \text{--- (ii)}$$

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x}{r}$$

$$\sin \theta \sec \phi = A(r, \theta, \phi) \quad \text{--- (a)}$$

Differentiating (ii):

$$\sec^2 \theta \frac{d\theta}{dx} = \frac{dx}{y}$$

$$\Rightarrow \frac{\partial \theta}{\partial x} = \frac{\cos^2 \theta}{y} = \frac{\cos^2 \theta}{r \cos \theta \sec \phi}$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{r} \cos \theta \cos \phi = B(r, \theta, \phi) \quad \text{--- (b)}$$

Differentiating (iii):

$$\sec^2 \phi \frac{\partial \phi}{\partial x} = -\frac{z}{r^2} \frac{\partial r}{\partial x}$$

$$= -\frac{z}{r^2} \cdot \frac{x}{r} = -\frac{r \tan \phi \cdot r \sin \theta \sec \phi}{r^3}$$

$$\therefore \frac{\partial \phi}{\partial x} = -\frac{\tan \phi \sin \theta \sec \phi}{r}$$

$$\Rightarrow \sec^2 \phi \frac{\partial \phi}{\partial x} = -\frac{\tan \phi \sin \theta \sec \phi}{r}$$

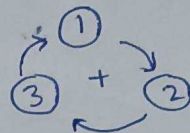
$$\Rightarrow \frac{\partial \phi}{\partial x} = -\frac{\sin \phi \sin \theta}{r} = C(r, \theta, \phi) \quad \text{--- (c)}$$

$$\therefore \frac{\partial}{\partial x} = \sin \theta \sec \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{1}{r} \sin \theta \sin \phi \frac{\partial}{\partial \phi}$$

(Ans)

(b) (A) $\epsilon_{213} = -1$ For $j=1$: $\epsilon_{213} = -1$ $\delta_{11} = 1$
 $\epsilon_{213} = -1$

FALSE



(B) For $j=1$: $\epsilon_{213} = -1$

$\delta_{12} \delta_{13} - \delta_{11} = -1$

TRUE

(C) For $j=2$: $\epsilon_{223} = 0$

$\delta_{21} + \delta_{23} - \delta_{21} = 0$

TRUE

For $j=3$: $\epsilon_{233} = 0$

$\delta_{32} \delta_{33} - \delta_{31} = 0$

(D) For $j=1$: $\epsilon_{213} = -1$ ~~$\delta_{21} - \delta_{13}$~~

$\delta_{21} - \delta_{23} = 0$

FALSE

(E) For $j=1$: $\epsilon_{213} = -1$; $-\delta_{11} = -1$

For $j=2$: $\epsilon_{223} = 0$; $-\delta_{21} = 0$

TRUE

For $j=3$: $\epsilon_{233} = 0$; $-\delta_{31} = 0$

(F) For $j=1$: $\epsilon_{213} = -1$

$1 - \delta_{11} = 0$

FALSE

(2) (a) Done previously in Midsem (82a)

(b) $(D-a)(D-b)x(t) = e^{ct}$ — (i)

Let $(D-b)x(t) = y(t)$ — (ii)

Then the equation (i) reduces to:

$(D-a)y(t) = e^{ct}$ — (iii)

From the definition of PD given above:

$$y(t) = e^{at} \int_0^t e^{ct'} e^{-at'} dt' \\ = e^{at} \int_0^t e^{(c-a)t'} dt' = e^{at} \left[\frac{e^{(c-a)t'}}{c-a} \right]_0^t$$

$$\Rightarrow y(t) = \frac{e^{at}}{c-a} [e^{(c-a)t} - 1] \quad (20)$$

$$\Rightarrow y(t) = \frac{e^{ct} - e^{at}}{c-a}$$

From eqn (1)

$$x(t) = e^{bt} \int_0^t y(t') e^{-bt'} dt'$$

$$= \frac{e^{bt}}{c-a} \int_0^t (e^{ct'} - e^{at'}) e^{-bt'} dt'$$

$$= \frac{e^{bt}}{c-a} \int_0^t e^{(c-b)t'} - e^{(a-b)t'} dt'$$

$$= \frac{e^{bt}}{c-a} \left[\frac{e^{(c-b)t'}}{c-b} - \frac{e^{(a-b)t'}}{a-b} \right]_0^t$$

$$\Rightarrow x(t) = \frac{e^{bt}}{c-a} \left[\frac{e^{(c-b)t}}{c-b} - \frac{e^{(a-b)t}}{a-b} \right] \quad (\text{Ans})$$

(3) Done previously in Midsem (Q3)

$$(4) \quad \dot{x} = \sec x \cos y \quad ; \quad \dot{y} = \sin x \operatorname{cosec} y$$

$$\text{Let } u = \sin x \quad ; \quad v = \cos y$$

$$\Rightarrow \dot{u} = \cos x \dot{x} \quad \Rightarrow \dot{v} = -\sin y \dot{y} \quad (i)$$

$$\dot{x} = \sec x v \quad \Rightarrow v = \dot{x} \cos x \quad (ii)$$

$$\dot{y} = \sin x \operatorname{cosec} y \quad \Rightarrow u = \dot{y} \sin y \quad (iv)$$

From (i), (ii), (iii) & (iv):

$$\dot{u} = v \quad \& \quad \dot{v} = -u$$

$$\therefore \frac{d}{dt} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\Rightarrow \frac{dx}{dt} = Mx$$

$$\Rightarrow x(t) = e^{Mt} x(0) \quad (v)$$

$$u(0) = \sin(x(0)) = \sin a$$

$$v(0) = \cos(y(0)) = \cos b$$

(21)

$$\therefore X(0) = \begin{bmatrix} u(0) \\ v(0) \end{bmatrix} = \begin{bmatrix} \sin a \\ \cos b \end{bmatrix} \quad \text{--- (vi)}$$

$$\therefore M^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$$

$$\therefore M^3 = M^2 M = -M$$

$$\therefore M^4 = (M^2)^2 = I$$

Powers of M behave like the powers of i .

$$\therefore e^{Mt} = I + \frac{Mt}{1!} + \frac{(Mt)^2}{2!} + \frac{(Mt)^3}{3!} + \dots$$

$$= I + \frac{Mt}{1!} + \frac{I t^2}{2!} + \frac{M t^3}{3!} + \dots$$

$$= I \left(1 - \frac{t^2}{2!} + \dots \right) + M \left(t - \frac{t^3}{3!} + \dots \right)$$

$$\Rightarrow \boxed{e^{Mt} = I \cos t + M \sin t} \quad \text{--- (vii)}$$

From (vi), (vii) & (viii):

$$X(t) = e^{Mt} = \begin{bmatrix} \cos t & 0 \\ 0 & \cos t \end{bmatrix} + \begin{bmatrix} 0 & \sin t \\ -\sin t & 0 \end{bmatrix}$$

$$\boxed{e^{Mt} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}} \quad \text{--- (viii)}$$

Substituting (vi) & (viii) into (v):

$$X(t) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} \begin{bmatrix} \sin a \\ \cos b \end{bmatrix}$$

$$= \begin{bmatrix} \sin a \cos t + \sin t \cos b \\ -\sin t \sin a + \cos b \cos t \end{bmatrix} = \begin{bmatrix} u(t) \\ v(t) \end{bmatrix}$$

$$= \begin{bmatrix} \sin(a+t) \\ \cos(a+t) \end{bmatrix}$$

$$\Rightarrow u(t) = \sin a \cos t + \sin t \cos b$$

(22)

$$v(t) = -\sin a \sin t + \cos b \cos t$$

$$\therefore x(t) = \sin^{-1}(u(t))$$

$$\boxed{x(t) = \sin^{-1}(\sin a \cos t + \sin t \cos b)} \quad (\text{Ans})$$

$$\therefore y(t) = \cos^{-1}(v(t))$$

$$\boxed{y(t) = \cos^{-1}(\cos b \cos t - \sin a \sin t)} \quad (\text{Ans})$$

(5) (a) $M = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$

We need to diagonalise the matrix.

$$M = UDU^{-1}$$

$$\Rightarrow f(M) = U f(D) U^{-1}$$

Eigen values:

Let the eigenvalue of M be λ .

$$\therefore |M - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} a-\lambda & 0 \\ b & c-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (a-\lambda)(c-\lambda) = 0 \Rightarrow \boxed{\lambda = a \text{ \& \& } \lambda = c}$$

$$\therefore D = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix}$$

Eigenvectors:

(1) $\lambda = a$:

Let the eigenvector be $A = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

$$\therefore (M - aI) \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 \\ b & c-a \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$ba_1 + (c-a)a_2 = 0$$

$$\Rightarrow a_2 = -\frac{ba_1}{a-c}$$

Taking $a_1 = 1$; $a_2 = \frac{b}{a-c}$

$$A = \begin{pmatrix} 1 \\ \frac{b}{a-c} \end{pmatrix}$$

(ii) $\lambda = c$

Let the eigenvector be $B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

$$(M - cI) \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = 0$$

$$\begin{bmatrix} a-c & 0 \\ b & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = 0$$

$$(a-c)b_1 = 0 \Rightarrow b_1 = 0 \quad [\because a \neq c]$$

$$bb_2 = 0 \Rightarrow b_2 = 0$$

$$B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{taking } b_2 = 1)$$

$$U = (A \ B) = \begin{pmatrix} 1 & 0 \\ \frac{b}{a-c} & 1 \end{pmatrix} = U$$

$$U^{-1} = \begin{pmatrix} 1 & 0 \\ \frac{b}{-a+c} & 1 \end{pmatrix}$$

(exchange the diagonal elements & flip the sign of non-diagonal elements)

$$\sin^2 M = U \sin^2 D U^{-1}$$

$$= \begin{bmatrix} 1 & 0 \\ \frac{b}{a-c} & 1 \end{bmatrix} \begin{bmatrix} \sin^2 a & 0 \\ 0 & \sin^2 c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{b}{c-a} & 1 \end{bmatrix}$$

$$[D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}] \Rightarrow f(D) = \begin{pmatrix} f(\lambda_1) & 0 \\ 0 & f(\lambda_2) \end{pmatrix}$$

$$= \begin{bmatrix} \sin^2 a & 0 \\ \frac{b}{a-c} \sin^2 a & \sin^2 c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{b}{c-a} & 1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} \sin^2 a & 0 & 0 \\ \frac{b}{a-c} (\sin^2 a - \sin^2 c) & \sin^2 c \end{bmatrix} \quad (\text{Ans})$$

(24)

$$(b) A^n = \sin^{2n}(M)$$

$$= U \sin^{2n}(D) U^{-1}$$

$$= \begin{bmatrix} 1 & 0 \\ \frac{b}{a-c} & 1 \end{bmatrix} \begin{bmatrix} \sin^{2n} a & 0 \\ 0 & \sin^{2n} c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{b}{c-a} & 1 \end{bmatrix}$$

$$A^n = \begin{bmatrix} \sin^{2n} a & 0 \\ \frac{b}{a-c} (\sin^{2n} a - \sin^{2n} c) & \sin^{2n} c \end{bmatrix}$$

$\therefore \lim_{n \rightarrow \infty} A^n$ for any integer a , $0 \leq \sin^2 a < 1$
 As $n \rightarrow \infty$, $(\sin^2 a)^n = 0$

$$\boxed{\lim_{n \rightarrow \infty} A^n = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}} \quad \text{Ans}$$